Realised Volatility Forecasting — Multi-Asset Benchmark

Direct Multi-Step Models: HAR, NLinear, DLinear, SAS, RNN

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1 Setup

$x_t = \log \mathrm{RV}_t$, z-scored per rolling window (μ, σ frozen until refit). Direct multi-step: one model per $h \in \{1, 5, 22\}$. Ridge $\alpha \in \{10^k : k \in [-4, 5],\ \Delta k{=}0.25\}$ (37 values), 5-fold CV.

2 Models

2.1 HAR — Heterogeneous Autoregressive Model (Corsi 2009)

Features. Let $\bar{x}_t^{(w)} = \frac{1}{5}\sum_{k=0}^{4} x_{t-k}$ and $\bar{x}_t^{(m)} = \frac{1}{22}\sum_{k=0}^{21} x_{t-k}$.

$$\hat{x}_{t+h} = \alpha + \beta_d\, x_t + \beta_w\, \bar{x}_t^{(w)} + \beta_m\, \bar{x}_t^{(m)} \tag{1}$$

HAR-OLS: $\min_\beta \|X\beta - y\|_2^2$. **HAR-Ridge**: column-normalise $[x^{(d)}, x^{(w)}, x^{(m)}]$; penalise slopes only; back-transform $w = w_{\text{scaled}}/\sigma_{\text{col}}$.

2.2 NLinear and DLinear (Zeng et al. 2023)

Let $\boldsymbol{x}_t = (x_{t-L+1}, \ldots, x_t)^\top \in \mathbb{R}^L$ be the lookback window.

NLinear.

$$\tilde{\boldsymbol{x}}_t = \boldsymbol{x}_t - x_t, \qquad \hat{x}_{t+h} = \tilde{\boldsymbol{x}}_t^\top w_h + x_t \tag{2}$$

DLinear.

$$\boldsymbol{\tau}_t = \mathrm{MA}_k(\boldsymbol{x}_t), \quad \boldsymbol{\varsigma}_t = \boldsymbol{x}_t - \boldsymbol{\tau}_t, \qquad \hat{x}_{t+h} = \begin{bmatrix} \boldsymbol{\tau}_t \\ \boldsymbol{\varsigma}_t \end{bmatrix}^\top w_h \tag{3}$$

2.3 GARCH — AR(p)-GARCH(1,1)

$x_t = \mu + \sum_{i=1}^{p} \phi_i x_{t-i} + \varepsilon_t, \quad \sigma_t^2 = \omega + \alpha_G \varepsilon_{t-1}^2 + \beta_G \sigma_{t-1}^2$. Multi-step: iterate variance recursion analytically. QMLE.

2.4 SAS — State Affine System

Diagonal basis. $P_0, \ldots, P_p \in \mathbb{R}^n,\ Q_0, \ldots, Q_q \in \mathbb{R}^n$.

$$P(z) = \mathrm{clip}\Big(\sum_{d=0}^{p} P_d\, z^d,\ -1{+}\epsilon,\ 1{-}\epsilon\Big) \in (-1,1)^n \quad \text{[Horner's rule]} \tag{4}$$

$$Q(z) = \sum_{d=0}^{q} Q_d\, z^d \in \mathbb{R}^n \tag{5}$$

$$s_t = P(z_t) \odot s_{t-1} + Q(z_t) \tag{6}$$

Clipping ensures $|P(z)_i| < 1$ (unconditional stability). Readout: $\hat{x}_{t+h} = s_t^\top w_h$.

Associative structure. $\ell_t = (P(z_t), Q(z_t))$. Composition rule:

$$(A_j,\, b_j) \circ (A_i,\, b_i) \;=\; \big(A_j \odot A_i,\;\; A_j \odot b_i + b_j\big) \tag{7}$$

is associative $\Rightarrow$ parallel divide-and-conquer scan.

Algorithm 1: Two-Level Parallel Scan — compute $s_1, \ldots, s_T$ from $z_1, \ldots, z_T$

Input : Transitions $\ell_1, \ldots, \ell_T$;;
chunk size B;;
$s_0 = \mathbf{0}$
Output : States $s_1, \ldots, s_T$;;
s_T

$K \leftarrow \lceil T/B \rceil$; pad to KB
Phase 1 (vmap over K chunks);
  $\forall\, k:\ (A^{\text{intra}}_{k,\cdot},\, b^{\text{intra}}_{k,\cdot}) \leftarrow \textsc{AssocScan}(\circ,\ \ell_{kB+1}, \ldots, \ell_{(k+1)B})$
Phase 2 (inter-chunk scan over K endpoints);
  $(A^{\text{inter}},\, b^{\text{inter}}) \leftarrow \textsc{AssocScan}(\circ,\ (A^{\text{intra}}_{1,B}, b^{\text{intra}}_{1,B}), \ldots)$
Phase 3 (carry for chunk $k \geq 1$);
  $c_0 = s_0$;;
$c_k = A^{\text{inter}}_{k-1} \odot s_0 + b^{\text{inter}}_{k-1}$
Phase 4 (vmap over K chunks);
  $\forall\, k, j:\ s_{kB+j} = A^{\text{intra}}_{k,j} \odot c_k + b^{\text{intra}}_{k,j}$

Depth $O(\log T)$ vs. $O(T)$ sequential.

Augmented SAS (AugSAS).

$$f_t = \big[s_t;\ 1,\ x_t,\ \bar{x}_t^{(w)},\ \bar{x}_t^{(m)}\big] \in \mathbb{R}^{n+4}, \qquad \hat{x}_{t+h} = f_t^\top w_h \tag{8}$$

$w_h[1{:}n] = \mathbf{0}$ recovers HAR $\Rightarrow$ AugSAS $\geq$ HAR by construction.

2.5 RNN — LSTM / GRU / LRU

Readout: $\hat{x}_{t+h} = \mathrm{Linear}_h(h_T)$, one head per h. Training: Adam, early stopping (patience=20, val=20% chronological).

LRU (Orvieto et al. 2023):

$$h_t = \sigma(\nu) \odot h_{t-1} + B z_t, \qquad y_t = C h_t, \qquad \sigma(\nu) \in (0,1)^d \tag{9}$$

3 Evaluation Protocol

Algorithm 2: Rolling Out-of-Sample Evaluation

Input : Series $x_1, \ldots, x_T$; window W; refit frequency R; horizons $\mathcal{H}$

Output : Squared-error sequences $\{e_{t,m}^{(h)}\}$ for each model m, horizon h

```
(μ, σ) ← (0, 1);
for t = W to T − max(H) do
    train ← x_{t−W:t};
    if t mod R = 0 then
        (μ, σ) ← FitScaler(train);
        traiñ ← (train − μ)/σ;
        foreach model m do
            m.Fit(traiñ, H)
    z_t ← (x_t − μ)/σ;
    foreach model m, horizon h ∈ H do
        ẑ_{t+h} ← m.Predict(h);
        e^{(h)}_{t,m} ← (ẑ_{t+h} − z_{t+h})^2;
    foreach model m do
        m.Update(z_t)
```

Model Confidence Set (Hansen et al. 2011). $L \in \mathbb{R}^{T \times M}$, row-normalised by cross-model mean to prevent high-vol dominance.

1. $d_{t,m} = L_{t,m} - \bar{L}_t$
2. $T_R = \max_m \bar{d}_m / \hat{\sigma}_m$ ($\hat{\sigma}_m$: circular block bootstrap, B reps)
3. Eliminate worst model if $p < \alpha$; repeat until H_0 not rejected.
4. Surviving set = MCS_α.

MCS Frequency. Run MCS per symbol per horizon; report $c_m^{(h)}/N$.

4 Hyperparameters

5 Results

5.1 Main Benchmark — All Models, 30 Symbols (W=2000, R=20)

5.2 SAS Hyperparameter Sweep — MCS Frequency, 30 Symbols (W=2000, R=20)

Comparison set: SAS family + HAR only. Higher frequencies than Table 3 due to smaller, more homogeneous set.

Table 1: Fixed hyperparameters used in all experiments.

Component	**Parameter**	**Value**
Evaluation	Training window W	2 000
	Refit frequency R	20 steps (≈1 month)
	Horizons $\mathcal{H}$	$\{1, 5, 22\}$ days
MCS	Significance α	0.10
	Bootstrap replications B	1 000
HAR	Lags (d, w, m)	$(1, 5, 22)$ days
	Ridge CV folds	5
NLinear / DLinear	Lookback L	20
	DLinear MA kernel	5
GARCH	AR lags p	5
	Estimation	QMLE
SAS (diagonal)	Reservoir size n	100 or 200
	Poly degree (p, q)	$(1, 1)$ or $(2, 2)$
	Spectral norm constraint	$\{0.80, 0.90, 0.95, 0.99\}$
	Washout	50 steps
RNN (GRU)	Hidden size	32
	Layers	1
	Lookback	20
	Epochs (max)	500
	Early stopping	patience = 20, val = 20%
Ridge grid	Range	10^k, $k \in [-4, 5]$
	Spacing	$\Delta k = 0.25$ (37 values)

Table 2: Out-of-sample MSE (mean ± std, 30 Oxford-Man symbols, target log-RV_5, sorted by Avg).

Model	$h = 1$	$h = 5$	$h = 22$	**Avg**
HAR	0.400 ± 0.072	0.510 ± 0.094	**0.689 ± 0.138**	**0.533**
HAR-Ridge	0.400 ± 0.072	0.510 ± 0.094	0.692 ± 0.143	0.534
$\mathbf{SAS_{diag}}$	**0.397 ± 0.071**	**0.508 ± 0.093**	0.708 ± 0.164	0.537
DLinear	0.400 ± 0.072	0.512 ± 0.094	0.700 ± 0.148	0.538
NLinear	0.403 ± 0.074	0.520 ± 0.098	0.717 ± 0.151	0.547
SAS_{aug}	0.402 ± 0.070	0.514 ± 0.096	0.724 ± 0.160	0.547
GARCH	0.410 ± 0.074	0.546 ± 0.098	0.831 ± 0.188	0.596

SAS_{diag}: `diag_n200_sn0.95`; SAS_{aug}: `aug_n100_sn0.90`

Table 3: $MCS_{0.10}$ frequency (row-normalised losses): #symbols where model survived the best set. Sorted by Total/90.

Model	$h = 1$	$h = 5$	$h = 22$	**Total/90**	**Avg MSE**
HAR	27/30 (90%)	29/30 (97%)	30/30 (100%)	86	0.533
HAR-Ridge	27/30 (90%)	24/30 (80%)	27/30 (90%)	78	0.534
SAS_{aug}	28/30 (93%)	17/30 (57%)	7/30 (23%)	52	0.547
SAS_{diag}	23/30 (77%)	16/30 (53%)	7/30 (23%)	46	0.537
DLinear	11/30 (37%)	5/30 (17%)	3/30 (10%)	19	0.538
NLinear	0/30 (0%)	0/30 (0%)	0/30 (0%)	0	0.547
GARCH	0/30 (0%)	0/30 (0%)	0/30 (0%)	0	0.596

Table 4: SAS sweep: $\mathrm{MCS}_{0.10}$ frequency (30 symbols) and mean MSE. ‡ = CV>0.5 on ≥1 horizon. **Bold** = configs used in main benchmark.

Model	$h = 1$	$h = 5$	$h = 22$	**Total**	**Avg MSE**
diag n=200**,** ρ=0.95	30/30	29/30	30/30	89	0.537
aug n=100**,** ρ=0.90	30/30	30/30	27/30	87	0.547
aug n=100, ρ=0.95	30/30	30/30	29/30	89	0.557
diag n=200, ρ=0.90	30/30	30/30	27/30	87	0.564
diag n=100, ρ=0.90	29/30	30/30	26/30	85	0.572
diag n=100, ρ=0.99	20/30	20/30	18/30	58	—
HAR-Ridge	5/30	14/30	26/30	45	0.534
HAR	5/30	14/30	25/30	44	0.533
diag n=100, ρ=0.95 ‡	30/30	30/30	30/30	90	0.582
diag p=2, ρ=0.95	14/30	16/30	7/30	37	0.560
diag p=2, ρ=0.90	8/30	13/30	8/30	29	0.572
diag n=100, ρ=0.80	12/30	9/30	5/30	26	0.649